

Use-Case Flow Document

Multi-Cloud Asset Cost Optimization Portal

Use Case Name	Monitor Budget & Send Breach Alert
Use Case ID	UC-03 (FR-003)
Primary Actor	System (scheduled job) / FinOps Analyst
Secondary Actor(s)	Cloud Administrator (receives alert, may adjust threshold)
Trigger	Scheduled cost-evaluation job runs (every 15 minutes) after billing data has been ingested
Preconditions	Billing data for the current period has been ingested (includes Ingest Cloud Billing Data); at least one project/team budget threshold is configured
Postconditions (Success)	A breach event is logged; the FinOps Analyst and Cloud Administrator are notified with the correct project, threshold, and current spend

Main Success Scenario

1. The scheduled cost-evaluation job retrieves the latest ingested billing data for each active project/team budget.
2. System calculates actual spend-to-date and a forecasted month-end spend for each project.
3. System compares actual and forecasted spend against the project's configured budget threshold.
4. System detects that actual or forecasted spend exceeds the threshold for a project.
5. System logs a budget-breach event containing the project ID, threshold, and current spend.
6. System dispatches an alert (email and in-app notification) to the FinOps Analyst and Cloud Administrator within 15 minutes of detection.
7. FinOps Analyst opens the alert and reviews the breach details on the cost dashboard.
8. Use case ends successfully.

Alternate Flow A1 — Notification Delivery Fails

A1.1. At step 6, the email or in-app notification service is unreachable and delivery fails.

A1.2. System retries delivery up to 3 times with exponential backoff.

A1.3. If delivery still fails, the system keeps the breach event flagged as “unacknowledged” and surfaces it prominently on the dashboard's alert banner.

A1.4. System logs the delivery failure for audit purposes; the use case continues at step 7 once the analyst next opens the dashboard.

Exception Flow E1 — No Budget Threshold Configured

E1.1. At step 3, the system finds that the project has no budget threshold configured.

E1.2. System skips breach evaluation for that project and logs it under “unmonitored projects” on the dashboard.

E1.3. System prompts the Cloud Administrator, on next login, to configure a threshold for the unmonitored project (extends Configure Alert Thresholds).

E1.4. Use case ends without a breach alert for that project.

Post-conditions (Failure)

- No alert is confirmed delivered; the breach remains visible as unacknowledged on the dashboard (A1).
- The project remains unmonitored until a threshold is configured (E1).